SS2 Part D – Business Recommendations

STADIOcare Group | Public-data proof of concept | 30-day readmission prediction

Executive recommendation

The public-data experiment supports further investigation of 30-day hospital readmission prediction, but not immediate clinical deployment. Both models distinguish risk better than random chance, yet neither offers strong discrimination or precision. Logistic Regression achieved ROC-AUC 0.641 and average precision 0.209; Random Forest achieved ROC-AUC 0.640 and average precision 0.198. At a fixed probability threshold of 0.5, Random Forest attained higher accuracy (74.0%) but Logistic Regression identified more actual early readmissions (recall 51.8% versus 40.2%). A phased validation programme is recommended rather than adoption of either model as a clinical rule.

Preferred model and rationale

Logistic Regression is the provisional preferred baseline for the next phase. Its recall was 51.8%, compared with 40.2% for Random Forest, and its ROC-AUC was marginally higher. This matters where missing a genuinely high-risk patient could mean a missed opportunity for post-discharge follow-up. Logistic Regression is also comparatively straightforward to explain to clinical managers. Nevertheless, its precision was only 17.0%; many patients flagged positive would not experience an early readmission. Random Forest's precision was 19.5%, also low. Selection should therefore be revisited using STADIOcare data, explicit clinical priorities and the costs of unnecessary follow-up versus missed readmissions. These observed results are not evidence that one model is statistically superior.

Improving model quality

First, define the prediction time: for example, immediately before discharge, using only information available by that point. Review every feature for leakage and clinical availability. Second, examine missingness and compare alternative handling of laboratory and specialty information rather than automatically excluding those fields. Third, optimise model hyperparameters and probability thresholds using patient-grouped cross-validation within the training data; retain an untouched test set. Fourth, evaluate calibration, sensitivity at practical alert volumes, positive predictive value, and subgroup performance across hospitals and demographic groups. Bootstrap confidence intervals would help quantify uncertainty. Finally, consider more expressive models only if they deliver a meaningful, reproducible improvement over the interpretable baseline. This experiment did not undertake tuning, calibration, external validation or statistical significance testing.

Adaptation to the SS1 data request

The SS1 proposal requested de-identified admission and patient identifiers, admission/discharge times, diagnosis and procedure codes, comorbidity indicators, prior hospital and emergency utilisation, discharge destinations and follow-up details. These should be assembled into a documented, admission-level analytical dataset. A consistent definition of unplanned readmission within 30 days is essential, including rules for transfers, planned procedures, deaths and patients readmitted to a different facility. Data linkage must be authorised, access controlled and audited. The public UCI data include broad analogues for age, diagnoses, prior utilisation, medications

and length of stay, but do not supply all STADIOcare variables, notably follow-up attendance and ICU history. Those additional variables may improve local relevance if collected reliably and available at prediction time.

Operational implementation and governance

Begin with a retrospective validation using several years of STADIOcare data and a time-based holdout that reflects future use. Then conduct a prospective silent trial in which scores are calculated but not used to change care. Assess whether alerts would be actionable by discharge coordinators, nursing teams and clinicians, and whether follow-up capacity is sufficient. Monitor calibration and performance over time, by hospital and patient subgroup, because case mix and care pathways may change. Clinical governance should specify accountability, data protection, model versioning, incident escalation, audit trails and human review. The system should support, not replace, clinical judgement.

Alignment with literature and conclusion

Strack et al. (2014) established the usefulness and limitations of structured diabetes encounter records for readmission research. Emi-Johnson and Nkrumah (2025) reported only moderate discrimination across several machine-learning models on the same public dataset, while Shang et al. (2021) demonstrated sensitivity of model performance to preprocessing and sampling choices. The present ROC-AUC values are broadly consistent with the modest discrimination reported in that literature, but direct numerical comparisons are limited by differences in patient selection, splitting and feature preparation. STADIOcare should proceed to local data-quality assessment and validation, with a clearly defined clinical intervention and measurable success criteria, before considering a pilot that affects patient care.

The dataset also creates important transferability limits. It represents diabetes-related inpatient encounters from US hospitals between 1999 and 2008, whereas STADIOcare operates in a different healthcare system and time period. Changes in treatment pathways, hospital coding, population characteristics and post-discharge services could alter which variables are useful. Even a well-performing model on the public dataset could fail locally. Before any clinical implementation, STADIOcare should confirm that definitions, coding practices and data capture are consistent across its hospitals. Missingness, duplicate encounters and the ability to track readmissions across facilities should be audited. If outcome capture is incomplete, apparently accurate predictions may simply reflect gaps in record linkage. These checks are prerequisites to trustworthy evaluation, not optional technical refinements.

Implementation should be judged against a measurable operational baseline. Before a live pilot, the organisation should record its current 30-day readmission rate, the proportion of eligible patients receiving timely follow-up, available case-management capacity and the time staff spend reviewing potential high-risk cases. A model that flags too many patients could create additional workload without improving outcomes. For this reason, a decision threshold should be chosen with clinicians and operational managers, based on how many patients the team can realistically contact and the expected consequences of false alarms and missed cases. A controlled pilot could compare follow-up completion and readmission outcomes against established practice, while recognising that observational changes do not necessarily prove a causal benefit.

References

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